The difference $\vec{A}-\vec{B}$ is the same as the resultant of $\vec{A}$ and $-\vec{B}$.

We can write $\vec{A}-\vec{B}=\vec{A}+\left(-\vec{B}\right)$ and using the equation (2.1), we have

$$\left|\vec{A}-\vec{B}\right|=\sqrt{A^2+B^2+2AB\cos(180-\theta)} \quad (2.3)$$

Since, $\cos\left(180-\theta\right)=-\cos\theta$, we get

$$\Rightarrow\left|\vec{A}-\vec{B}\right|=\sqrt{A^2+B^2-2AB\cos\theta} \quad (2.4)$$

Again from the Figure 2.19, and using an equation similar to equation (2.2) we have

$$\tan\alpha_2=\frac{B\sin\left(180^\circ-\theta\right)}{A+B\cos\left(180^\circ-\theta\right)} \quad (2.5)$$

But $\sin\left(180^\circ-\theta\right)=\sin\theta$ hence we get

$$\Rightarrow\tan\alpha_2=\frac{B\sin\theta}{A-B\cos\theta} \quad (2.6)$$

Thus the difference $\vec{A}-\vec{B}$ is a vector with magnitude and direction given by equations 2.4 and 2.6 respectively.

EXAMPLE 2.2

Two vectors $\vec{A}$ and $\vec{B}$ of magnitude 5 units and 7 units make an angle 60° with each other. Find the magnitude of the difference vector $\vec{A}-\vec{B}$ and its direction with respect to the vector $\vec{A}$.

Solution

Using the equation (2.4),

$$\left|\vec{A}-\vec{B}\right|=\sqrt{5^2+7^2-2\times5\times7\cos60^\circ}$$

$$=\sqrt{25+49-35}=\sqrt{39}\ \textit{units}$$

The angle that $\vec{A}-\vec{B}$ makes with the vector $\vec{A}$ is given by

$$\tan\alpha_2=\frac{7\sin60^\circ}{5-7\cos60^\circ}=\frac{7\sqrt{3}}{10-7}=\frac{7}{\sqrt{3}}=4.041$$

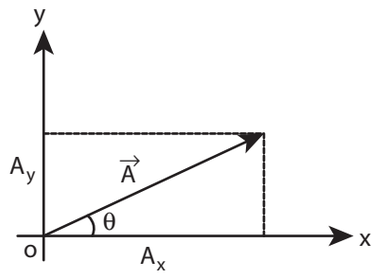
$$\alpha_2=\tan^{-1}\left(4.041\right)\cong76^\circ$$

2.4 COMPONENTS OF A VECTOR

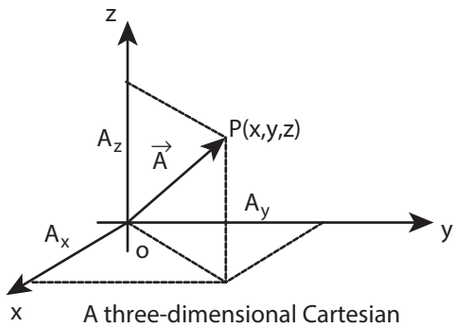
In the Cartesian coordinate system any vector $\vec{A}$ can be resolved into three components along *x*, *y* and *z* directions. This is shown in Figure 2.20.

Consider a 3–dimensional coordinate system. With respect to this a vector can be written in component form as

$$\vec{A}=A_x\hat{i}+A_y\hat{j}+A_z\hat{k}$$



A two-dimensional Cartesian coordinate system



A three-dimensional Cartesian coordinate system

Figure 2.20 Components of a vector in 2 dimensions and 3 dimensions

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